When personalization deters delegation to algorithms: Evidence from a preregistered experiment*

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Abstract

Algorithmic decision aids increasingly act on users' behalf, and in many instances users can either retain control or delegate decisions to an algorithm altogether. A common design assumption is that personalizing an algorithm to users' own preferences increases acceptance. Drawing on accounts of the intrinsic value of decision rights, we argue the opposite can hold: a rule that claims to decide as the user would substitutes for the user's own judgment, making the loss of choice authority salient at the moment of delegation. In a preregistered between-subjects experiment (N = 233), participants repeatedly chose whether to decide themselves or to delegate to an algorithm that selected the option with the highest expected value (in one treatment) or implemented participants' previously elicited preferences (in the other). Personalization did not increase delegation at any point of the task; instead, it deterred delegation in two ways. At first contact, participants were significantly less likely to delegate to the preference-based than to the optimization-based algorithm; in subsequent decisions, delegation rates were statistically indistinguishable. Moreover, in an exploratory analysis, categorical non-delegation (never delegating) was substantially more prevalent under personalization (15.7% vs. 3.4%). This resistance does not appear to be accuracy-based: participants expected the personalized algorithm to match their preferences *better* than the objective rule, yet were no more willing to delegate to it, and never-delegators emphasized retaining responsibility for the choice. Accuracy-based trust and authority-based resistance thus emerge as distinct channels of algorithm acceptance: personalization can improve the former while aggravating the latter.

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1 Introduction

Algorithmic systems increasingly make and implement decisions on users' behalf in both everyday and high-stakes contexts, from automated driving to autonomous trading and smart-home control. Public perceptions of algorithmic decision-making vary substantially across domains and are shaped by perceived fairness, accountability, privacy, trust, and usefulness (Aysolmaz, Müller, & Meacham, 2023). In many such settings, users face a concrete choice: they can retain control and decide themselves, or delegate the decision to an algorithm. Delegation can reduce cognitive effort and simplify complex choice tasks, but it can also reshape perceived agency and responsibility for outcomes. Recent literature shows that people do delegate consequential choices to AI systems, and that this delegation depends on the decision context (Candrian & Scherer, 2022). Note that throughout the paper, we refer to these systems interchangeably as algorithms, algorithmic advice, and algorithmic decision aids. This reflects the practical reality that, from a user's perspective, the boundary between AI and algorithmic decision rules is often opaque, and particularly so when we think about personalized systems.

The idea of personalizing algorithmic systems is to tailor content, products, and services to individual user preferences, which in turn should increase acceptance. Specifically, if individuals differ in preferences, then a decision aid that incorporates those preferences should, in principle, deliver higher subjective utility than a one-size-fits-all rule. In financial decision support, for example, systems often elicit risk preferences and translate them into portfolio recommendations. The key empirical question, however, is not whether personalization sounds attractive in principle, but whether it actually increases willingness to delegate.

Existing evidence leaves that question open. In robo-advisor research, much of the evidence focuses on what predicts whether people adopt robo-advisors and how they use them (for example, Mahmud, Islam, & Krishnan, 2025), rather than on direct contrasts between preference-based personalization and objective delegation rules. Adjacent work has shown that conversational onboarding and autonomy-preserving designs shape trust and uptake (Fink, Newman, & Haran, 2024; Hildebrand & Bergner, 2021). However, these studies vary the interface, degree of user control, or communication format, not the nature of the underlying decision rule itself. To our knowledge, no study has experimentally tested whether making an algorithm's decision rule preference-based, rather

than objectively optimizing, changes actual delegation behavior. More generally, behavioral evidence shows that delegation to algorithms is shaped by more than expected performance: people sometimes avoid algorithms and sometimes prefer them based on the context (Dietvorst, Simmons, & Massey, 2018; Logg, Minson, & Moore, 2019); they need to understand how recommendations are generated before accepting them (Yeomans, Shah, Mullainathan, & Kleinberg, 2019); and acceptance depends on moderators such as perceived control and task features (Chugunova & Sele, 2022; Mahmud, Islam, Ahmed, & Smolander, 2022).

Preference-based personalization can therefore be double-edged. While it may increase perceived fit, it may also encroach on users' sense of choice authority: a system that claims to decide *as you would* substitutes for personal judgment more directly than an impersonal optimization rule, making delegation feel like ceding active control rather than outsourcing a computation. Our own post-task evidence is consistent with this interpretation: many categorical non-delegators explicitly report wanting to keep responsibility for the choice, suggesting that the barrier is not fear of being held responsible for a bad choice or outcome, but rather a positive valuation of exercising choice authority. This reading aligns with evidence that perceived control over one's choices is a key driver of algorithm acceptance (Schaap, Bosse, & Hendriks Vettehen, 2024) on top of expected benefits. It also connects to findings that algorithmic transparency can paradoxically heighten resistance: disclosing how a system decides can trigger negative social interpretations (Chen, Dang, & Liu, 2024) and induce pushback in algorithmic management contexts (Hu, Zeng, Wang, & Teng, 2024), suggesting that transparency about a system's decision basis may itself contribute to a heightened sense of reduced agency.

These issues are especially salient at first contact, when users have no experiential basis for evaluating an algorithm and must judge based on abstract cues about its decision logic. Stable preferences for decision autonomy can constrain delegation even when delegation is instrumentally attractive (Ivanova-Stenzel & Tolksdorf, 2025), making the perceived loss of choice authority especially salient at the moment of delegation. Task characteristics matter too: trust and reliance depend on perceived objectivity and time pressure (Yang et al., 2025). Skepticism does not only operate before first contact, however: once delegation has begun, reactions to algorithmic errors are an important driver of whether users keep relying on a system. A robust finding is that people abandon algorithms more readily than they switch away from doing tasks themselves or delegating to another human after seeing comparable mistakes: algorithmic errors are, in effect, forgiven less (Dietvorst, Simmons, & Massey, 2015; Renier, Schmid Mast, & Bekbergenova, 2021). We will see that this asymmetry shapes how participants respond to feedback in our setting as well. Together, these factors suggest substantial skepticism about delegating to algorithms, both at the point of adoption

and in the face of subsequent errors.

To understand what shapes these first-contact decisions, we must study observable behavior rather than attitudes alone. Behavioral reliance and self-reported trust need not coincide; evidence from AI-advisor interactions shows that determinants of behavioral trust can diverge from self-reported trust (Schreibelmayr & Mara, 2025). Moreover, even in repeated interaction settings, the mere framing of advice as AI-generated can produce systematic overreliance (Klingbeil, Grützner, & Schreck, 2024), underscoring that label-driven cues persist beyond first contact. This motivates our focus on observable delegation behavior, treating first contact as a critical moment when abstract cues about an algorithm's decision logic are especially influential in shaping whether users delegate.

In this study, we contrast two decision rules that are both plausible in financial choice settings but may elicit different reactions. One algorithm follows an *objective optimization rule* and selects the option with the highest expected value (EV), which may appear impersonal and easier to justify as a rule-based benchmark. The other is *preference-based personalization* in a narrow sense: it predicts a participant's preferred choice from their previously elicited preferences rather than maximizing EV. In implementation terms, the personalized algorithm maps each participant's earlier choices in a simpler elicitation environment to corresponding choices in a more complex environment later in the experiment; it does not involve adaptive updating or learning during the delegation task.

To test this contrast, we run a preregistered between-subjects experiment with repeated delegation decisions. Participants first complete 12 binary lottery choices (Part 1), which provide the preference-elicitation input for the personalized algorithm. They then enter Part 2, where in each round they choose either to decide themselves among eight lottery options or to delegate that choice to the assigned algorithm (EV or preference-based, depending on the treatment).

This design allows us to isolate a preregistered first-contact test, before any outcome feedback is observed, and to distinguish it from behavior after feedback has accumulated. The comparison between our algorithms connects to evidence that responsibility-sharing considerations can shape advisor choice in human-versus-algorithm settings (Gazit, Arazy, & Hertz, 2023) and to recent studies of financial delegation and outcome evaluation (Ismagilova & Ploner, 2025). Related findings in financial delegation are also suggestive of responsibility-sharing considerations (Werner, Suraci, & Wolff, 2026), even though evidence on perceived responsibility differences between human and machine partners is mixed (Kirchkamp & Strobel, 2019).

Theoretical framework: delegation as ceding choice authority

Why should the content of an algorithm's decision rule change willingness to delegate, holding constant that delegation is available, voluntary, and instrumentally attractive? Our answer starts from the observation that choosing is not only a means to outcomes but carries value of its own. People derive procedural utility from being the agent who decides (Frey, Benz, & Stutzer, 2004), hold an intrinsic valuation of decision rights that exceeds their instrumental value (Bartling, Fehr, & Herz, 2014), and pay a measurable premium to retain control over payoff-relevant choices even when ceding control would yield better expected outcomes (Owens, Grossman, & Fackler, 2014). The desire for control appears to be a basic psychological need with deep biological roots (Leotti, Iyengar, & Ochsner, 2010), and its frustration can trigger reactance, a motivational state aimed at restoring threatened freedom of choice (Brehm, 1966). On this view, delegation to an algorithm is a transfer-of-control decision (Candrian & Scherer, 2022) whose psychological cost is not exhausted by expected outcomes: it includes the forgone value of exercising choice authority.

The decision rule matters because it shapes how this transfer of control is construed. Delegating to an impersonal optimization rule can be framed as *outsourcing a computation*: the algorithm performs an objective calculation the user could not (or would not) perform, and the user's judgment remains conceptually intact, much like using a calculator does not threaten one's sense of being the decision-maker. Delegating to a preference-based rule, by contrast, is harder to construe this way. A system that claims to decide *as the user would* does not complement the user's judgment; it replaces it. If the algorithm already chooses what the user would choose, the act of choosing itself becomes redundant, and it is precisely this act that carries intrinsic value (Bartling et al., 2014; Owens et al., 2014). Personalization thus makes the loss of choice authority salient in a way that objective optimization does not, threatening autonomy as a basic need (Deci & Ryan, 2000) and potentially evoking reactance (Brehm, 1966). This construal-based account is consistent with evidence that perceived control, rather than the algorithmic label per se, drives acceptance (Schaap et al., 2024), and that stable preferences for decision autonomy constrain delegation even when delegation pays (Ivanova-Stenzel & Tolksdorf, 2025).

A lottery-choice paradigm suits this construal mechanism because it lets us equate the two algorithms on everything except the feature our theory targets. Payoffs and probabilities are stated unambiguously, so perceived competence cannot be confounded with domain expertise. Preferences are directly elicitable in Part 1 and thus known to the experimenter, and expected value is computable from the displayed payoff table, so the two rules can be equally simple to state and equally legitimate as decision procedures, differing mainly in whether the

optimized criterion is impersonal or tied to the user's own profile. Real financial or medical decisions could not deliver this contrast cleanly: outcomes are ambiguous or deferred, preferences must be inferred from noisy self-report or past behavior, and the "correct" choice is unknown or contested, so reluctance to delegate could reflect distrust of the algorithm's competence rather than a reaction to *whose* criterion is being applied.

This framework yields three testable implications. First, at first contact (when no outcome feedback is available and construal of the decision rule is the dominant cue) delegation should be *lower* to a preference-based than to an objective algorithm, despite the former's normative appeal (our preregistered H1 below). Second, because the intrinsic valuation of decision rights is heterogeneous across individuals (Bartling et al., 2014; Owens et al., 2014), resistance should be concentrated in a subgroup: users who place high value on choice authority should opt out categorically, so categorical non-delegation should be more prevalent under personalization (a supplementary, non-preregistered implication we examine below). Third, the framework carries a mechanism signature that distinguishes it from accuracy-based accounts: resistance to the personalized algorithm should *not* be accompanied by lower expectations about its preference-matching accuracy. If users expect the personalized rule to match their preferences at least as well as (or better than) the objective rule, and yet delegate less, accuracy-based trust accounts become hard to sustain and authority-based accounts become the more plausible reading, although not the only possible one (see Limitations). Our post-task expectation measures let us examine this implication, with the caveat that they were collected after the incentivized rounds and concern preference matching rather than payoffs.

Preregistered hypothesis and analysis plan

Consistent with our preregistration (`https://aspredicted.org/7un5qf.pdf`), our main hypothesis is that the diminished sense of choice authority from delegating to a personalized algorithm outweighs the benefits it renders in terms of better-fitting choices:[3]

H 1. People are more likely to delegate financial decisions to an EV-maximizing algorithm than to a personalized algorithm that incorporates their (revealed) preferences.

Our preregistered primary analyses test this hypothesis in two complementary ways. First, we test the treatment difference in delegation in the first decision

[3]The direction of the prediction was informed by a pilot study with a different design (reported in Suraci's PhD thesis); see the Discussion section for a comparison of results.

(round 1) using a one-sided Boschloo exact test. This isolates first contact, when participants have no outcome feedback and must decide whether to delegate based on their interpretation of the algorithm. Second, we test whether EV maximization leads to higher delegation overall by comparing participants' average delegation frequencies across rounds using a one-sided Wilcoxon–Mann–Whitney test.

To account for the repeated-decision structure and additional determinants of delegation, our preregistration also specifies a multivariate regression analysis. We estimate a model with treatment indicators interacted with the decision period, decision-ID fixed effects, and participant-clustered standard errors. We further include the feedback history in the form of including the share of zero outcomes among prior delegated choices, to control for how experienced payoffs may influence subsequent delegation decisions. This preregistered specification tests whether any treatment difference varies systematically over decisions while controlling for option-set heterogeneity and participants' feedback histories. Finally, the preregistration contains one secondary hypothesis on feedback: after delegating to the algorithm and obtaining a payoff of zero, participants in the preference-based condition are more likely to switch to non-delegation in the following round than under any other circumstances, in particular than under the same circumstances in the EV condition. We report this test in the Results section.

Beyond our confirmatory test of H1, we exploratorily examine whether resistance to personalization is concentrated in a subgroup (categorical non-delegation) and whether authority-based and accuracy-based accounts can be dissociated using post-task expectation data.

Study overview and supplementary analyses

We implement our design in a between-subjects experiment in which participants repeatedly choose whether to decide themselves or delegate to the assigned algorithm. The repeated-decision setting provides a direct behavioral measure of delegation and allows us to distinguish first-contact delegation from later decisions.

In addition to the preregistered tests, we report a small set of analyses to characterize heterogeneity in delegation propensity. In particular, we examine categorical non-delegation (never delegating across rounds). This outcome captures a distinct form of resistance to algorithmic control transfer and is relevant for understanding which users may opt out of decision aids altogether.

In sum, this paper provides a preregistered test of a simple but consequential claim: even in a setting where preference-based personalization may appear normatively attractive, people may nonetheless be more willing to delegate to a 'more objective' algorithm. The contribution is threefold: we provide a preregistered

behavioral comparison of objective versus personalized decision rules and show that the treatment effect is concentrated at first contact; we document categorical non-delegation as a distinct uptake barrier under personalization; and we show that this resistance arises *despite* participants crediting the personalized algorithm with superior preference-matching, which is more consistent with an authority-based than with an accuracy-based barrier to algorithm adoption. Together, these findings inform both the theoretical understanding of algorithm aversion as a phenomenon with distinct accuracy-based and autonomy-based channels, and the design implication that uptake depends on perceived rule characteristics, not only on expected performance.

2 Related Work

Our work sits at the intersection of literatures on (i) the behavioral uptake of algorithmic decision aids, (ii) delegation, agency, and responsibility in human–algorithm interaction, (iii) transparency as an interpretive cue for acceptance, (iv) personalization as a design feature of algorithmic decision rules, and (v) heterogeneity in trust and delegation. Below, we situate our study within these streams, identify the specific gap our experiment addresses—the lack of a direct behavioral comparison between personalized and objective decision rules—and connect them to the robo-advisor application domain.

2.1 Algorithm aversion, appreciation, and conditions for uptake

Behavioral uptake of algorithmic advice is not one-directional: studies document both *algorithm aversion* and *algorithm appreciation*, depending on context, feedback, and comparison benchmark (Dietvorst et al., 2018; Logg et al., 2019). A core implication is that reliance cannot be predicted based on objective performance alone. Users need to make sense of how recommendations are generated before accepting them, and understandability can gate uptake independently of accuracy (Yeomans et al., 2019). Systematic evidence further shows that uptake is moderated by factors at multiple levels (algorithm, individual, and task), including perceived control and decision context (Chugunova & Sele, 2022; Mahmud et al., 2022).

This mixed-evidence baseline motivates our design choice to study delegation as an observed behavioral outcome and to isolate one concrete determinant of uptake: the algorithmic *decision rule* (objective EV maximization versus preference-based personalization).

2.2 Delegation, agency, and responsibility sharing

Delegation goes beyond advice-taking: it is a transfer-of-control decision that entails ceding choice authority to the algorithm (Candrian & Scherer, 2022). For that reason, agency and responsibility become central concerns. In advisor-choice settings, people evaluate more than expected payoff, including whether responsibility for consequences can be shared with or offloaded onto the advisor (Gazit et al., 2023). Related evidence from financial delegation suggests that delegation patterns are consistent with responsibility-sharing motivations, particularly in loss contexts (Werner et al., 2026). Yet, evidence on whether perceived responsibility differs between human and machine partners is mixed, with Kirchkamp and Strobel (2019) finding little difference in perceived responsibility or guilt.

Beyond responsibility considerations, a well-established body of evidence shows that people attach value to holding decision rights as such. Bartling et al. (2014) demonstrate that decision rights carry intrinsic value beyond their instrumental benefits; Owens et al. (2014) document a "control premium" whereby individuals sacrifice expected payoff to retain authority over choices; and Frey et al. (2004) argue that the process of deciding generates utility independently of outcomes. In the delegation-to-algorithms context specifically, Ivanova-Stenzel and Tolksdorf (2025) show that stable autonomy preferences constrain delegation even when it is instrumentally attractive. Our theoretical framework builds directly on this evidence: if choice authority is intrinsically valued, decision rules that make its loss more salient should face stronger resistance.

This control-and-responsibility perspective is consistent with adjacent evidence: perceived trustworthiness dimensions shape acceptance of algorithmic decision-makers (Höddinghaus, Sondern, & Hertel, 2021); preserving human involvement can increase uptake but may lower accuracy (Sele & Chugunova, 2024); and stable preferences for decision autonomy can constrain delegation even when delegation is instrumentally attractive (Ivanova-Stenzel & Tolksdorf, 2025). Taken together, this evidence implies that delegation is shaped by perceived control and responsibility allocation, not only expected returns. This motivates our focus on observed delegation behavior and makes the EV-versus-personalized comparison directly informative, because the two rules differ in how strongly they cue impersonal objectivity versus personal ownership of outcomes.

2.3 Transparency, and interpretive cues for acceptance

Personalization and transparency are often presented as acceptance-enhancing design features, but recent evidence suggests they can also trigger resistance. In recommendation contexts, greater transparency can backfire when users infer socially negative meanings (for example, feeling judged or dehumanized), reduc-

ing acceptance (Chen et al., 2024). In algorithmic management, transparency can likewise show a non-monotonic effect, where additional transparency eventually increases resistance rather than compliance (Hu et al., 2024). Related evidence further indicates that acceptance is driven less by whether the decision-maker is labeled algorithmic or human, and more by expected benefits and perceived control (Schaap et al., 2024).

The broader explainability literature is consistent with this interpretation: transparency and explanations matter, but their effects depend on what they change psychologically. Fairness/accountability/transparency perceptions shape trust and adoption (Shin & Park, 2019); explanations can improve trust calibration and behavior in some settings (Leichtmann, Humer, Hinterreiter, Streit, & Mara, 2023); and combining explanation with decision control can improve compliance, though design complexity can also offset those gains (Westphal, Vössing, Satzger, Yom-Tov, & Rafaeli, 2023). In applied decision-support experiments, transparency influences advice use via trusting beliefs and perceived understanding, with effects that depend on the information environment and experienced system performance (Ning, Lu, Li, & Gupta, 2024; Sun, Liao, Sundar, & Walther, 2026). For our setting, this implies that users' interpretation of decision logic can shape delegation independently of objective expected returns.

2.4 Decision-rule comparison gap: personalized versus objective rules

Despite the breadth of the literatures reviewed above, a striking gap remains: to our knowledge, no study has experimentally tested whether making an algorithm's decision rule *preference-based* (as opposed to following an impersonal, objective criterion) changes actual delegation behavior. Several lines of work come close but address different questions.

First, the "algorithm modification" paradigm pioneered by Dietvorst et al. (2018) shows that giving users even slight control over an algorithm's output increases adoption. This demonstrates that *user input* matters, but the manipulation concerns users' ability to adjust the output, not whether the underlying decision rule incorporates their preferences or maximizes an objective criterion.

Second, the robo-advisor design literature examines how features such as conversational onboarding (Hildebrand & Bergner, 2021), user controllability over rebalancing (S. Kim et al., 2025), and the performance–control trade-off in configuration choices (Rühr, 2020) shape adoption and trust. These studies vary the *interface* and the *degree of control*, but hold the nature of the decision rule itself constant or leave it as an implicit design feature.

Third, theoretical and empirical work on robo-advisor personalization has fo-

cused on optimizing portfolio allocation through more granular client interaction (Capponi, Olafsson, & Zariphopoulou, 2022) or on comparing algorithm-selected portfolios with self-directed choices (Boulu-Reshef, Direr, & von Wilczur, 2026). These contributions are informative about *performance* implications of personalization, but do not ask whether users are more or less willing to delegate when the algorithm's rule is personalized.

This gap is consequential beyond academic interest. In practice, financial robo-advisors, healthcare decision aids, and algorithmic recommender systems routinely face the design choice between objective optimization criteria (such as expected-value or evidence-based guidelines) and preference-based rules that incorporate individual user profiles. If personalization reduces rather than increases the willingness to delegate, as our evidence suggests, then system designers may inadvertently raise adoption barriers by prioritizing preference-matching. Understanding this trade-off is directly relevant for any domain where algorithmic decision aids must be voluntarily adopted to be effective.

2.5 Heterogeneity and contextual moderators of trust and delegation

Recent evidence shows that algorithm uptake is heterogeneous across people, tasks, and decision environments (Ferraz, Houf, Pitz, Schwieren, & Sickmann, 2025; Yang et al., 2025). At the individual level, delegation varies with psychological readiness and other personal traits linked to delegation behavior (Ferraz et al., 2025; H. Kim, 2026). Beyond stable traits, uptake also depends on decision architecture: autonomy-preserving choice designs can increase recommendation acceptance (Fink et al., 2024). At the task level, uptake depends on objectivity and time pressure (Yang et al., 2025), while overreliance risks remain salient when users follow advice against their own information (Klingbeil et al., 2024). Taken together, these cross-level moderators imply that average delegation rates can mask meaningful subgroup patterns. For our study, this motivates looking beyond means and explicitly reporting categorical non-delegation as a distinct acceptance pattern. The documented heterogeneity in the intrinsic valuation of decision rights (Bartling et al., 2014; Owens et al., 2014) provides a theoretical reason to expect exactly such a subgroup structure: if a minority of users values choice authority strongly, personalization-induced resistance could manifest as categorical opt-out in that subgroup rather than as a uniform shift in delegation rates.

2.6 Financial advice and robo-advisors as an application domain

Although our task is experimentally stylized, it maps naturally to financial delegation settings where users decide whether to retain control or hand choice authority to an algorithmic decision aid. This is exactly the context in which objective optimization and preference-based personalization often coexist in practice. Existing robo-advisor evidence is informative but fragmented: studies document adoption patterns and how users combine advisor recommendations with their own judgment (Figà-Talamanca, Tanzi, & D'Urzo, 2022; Mahmud et al., 2025), outcome bias in the evaluation of delegated financial decisions, whether made by a human or an algorithm (Ismagilova & Ploner, 2025), and potential performance gains from robo-advice relative to self-directed choices (Lambrecht, Oechssler, & Weidenholzer, 2026). This makes financial decision-making a strong test bed for our question: whether changing the algorithmic decision rule changes willingness to delegate, even when both rules are plausible.

2.7 Synthesis and research gap

Across these literatures, the common message is that delegation to algorithms is jointly shaped by decision-rule design, perceived control and responsibility, trust calibration, and context-dependent interpretation of transparency and autonomy-related design cues (Chugunova & Sele, 2022; Dietvorst et al., 2018; Fink et al., 2024; Hu et al., 2024; Schaap et al., 2024). Yet as we have documented above, the existing evidence base contains a notable gap: while adjacent work has studied algorithm modification, controllability, conversational design, and performance implications of personalized portfolios, no study has experimentally isolated the effect of the decision rule itself, specifically, whether making the rule preference-based rather than objectively optimizing changes users' willingness to delegate. This gap matters practically because robo-advisors, healthcare decision aids, and algorithmic recommender systems routinely face exactly this design choice, and a wrong assumption about the direction of the effect could raise rather than lower adoption barriers.

Our study addresses this gap with a preregistered behavioral test in a repeated-decision delegation paradigm, contrasting an EV-maximizing rule with a preference-based rule. It connects classic aversion/ appreciation evidence with newer findings on autonomy, transparency backfire, and responsibility attribution (Fink et al., 2024; Gazit et al., 2023; Hu et al., 2024; Logg et al., 2019), and anchors the predicted direction of the effect in the economics and psychology of choice authority (Bartling et al., 2014; Brehm, 1966; Deci & Ryan, 2000; Frey et al., 2004; Leotti et al., 2010; Owens et al., 2014). More broadly, our focus on the decision rule as a

determinant of delegation speaks to recent calls to move the study of human–AI interaction in finance from model performance toward decision architecture, and in particular, toward how authority, oversight, and accountability are allocated between human and algorithm (Kou, Li, Wang, & Wang, 2026); our results suggest that where this authority is perceived to reside can itself be shaped by how personalized the decision rule appears.

3 Method

3.1 Participants and preregistered exclusions

Participants were recruited via Prolific from a UK-based participant pool and completed the study online. The preregistered target was 120 participants per treatment condition before exclusions. We recruited N = 240 participants and then applied preregistered exclusion rules for data quality and protocol compliance. Exclusions covered failed bot/slider checks at first attempt, failed comprehension checks after the second attempt, failure to confirm careful participation, disclosed AI usage during the study, implausibly short completion times under the preregistered log-time rule, and incomplete data. The final analysis sample is N = 233 (EV condition: n = 118; preference-based condition: n = 115). Of the 7 excluded participants, 1 failed the bot/slider check, 4 failed the careful-participation check (one of whom also disclosed AI usage), and 3 disclosed AI usage during the study (one overlapping with the careful-participation failure); no participant was excluded for failing a comprehension check, for incomplete data, or under the log-completion-time rule.

3.2 Design and task

The study used a between-subjects design with two algorithm conditions. In both conditions, participants completed a two-part experiment programmed in the online-ready version of z-Tree (version 6.0.0 beta 69, Fischbacher, 2007).

Part 1 consisted of 12 binary lottery choices used to elicit revealed risk preferences. Eight of these tasks were directly linked to Part 2 (see below); the remaining four served as filler tasks, as one of several measures to prevent participants from anticipating the structural correspondence between Part 1 and Part 2. Participants were truthfully told that "[t]he questions in this part are **hypothetical**, but the choices **may be meaningful for Part 2** (recall that your bonus payment from the experiment depends on what happens in Part 2)." The highlighted words were boldfaced in the original instructions.

We deliberately designed Part 1 to give participants every opportunity to

make a well-considered, informed choice, precisely because these are the choices on which the preference-based algorithm is based. To this end, three features distinguish Part 1 from the later delegation task. First, unlike the time-constrained decisions in Part 2 (see below), participants faced no time limit in Part 1 and could deliberate for as long as they wished. Second, each Part-1 task presented a clean choice between only two lotteries, rather than the eight options participants would later face in Part 2. Third, payoffs were matched to (equally probable) states so that they would appear in decreasing order to ease comparison, as can be seen in the upper panel of Figure 1. In addition, we asked participants after each task in Part 1 how confident they were in their choice.

	State 1	State 2	State 3	State 4	State 5	State 6	State 7	State 8	State 9	State 10
Option 1	225	225	225	225	225	225	225	205	0	0
Option 2	525	525	525	525	525	0	0	0	0	0

State:	1	2	3	4	5	6	7	8	9	10
Option 1	225	225	0	225	0	225	225	225	225	0
Option 2	525	0	505	0	525	0	0	0	525	505
Option 3	525	0	525	0	525	0	0	0	525	525
Option 4	225	225	0	225	205	205	205	225	225	0
Option 5	225	205	0	205	205	225	225	205	225	0
Option 6	225	225	0	225	205	225	225	225	225	0
Option 7	525	0	525	0	525	0	0	0	225	525
Option 8	505	0	525	0	505	0	0	0	505	525

Figure 1: Example screens from Part 1 (preference elicitation, upper panel) and Part 2 (delegation decisions with lottery choices, lower panel).

In Part 2, participants completed eight incentivized decisions between eight lottery options each, as exemplified in the lower panel of Figure 1. These eight Part-2 tasks were derived from eight mapped Part-1 binary tasks (all but the four filler tasks). Specifically, each mapped Part-1 task provided two "anchor" options; for Part 2, we first transformed each anchor option by permuting payoffs across states (Option 6 and Option 3 in the lower panel of Figure 1), and then added three decoy options for each anchor option. Each added option was state-wise dominated by one of the anchor options (i.e., worse in at least one outcome and no better in any other), without making the dominance relationship immediately obvious from the displayed table. This yields eight options per Part-2 task (two transformed anchors plus six decoy options). The within-task display order of the eight options was set using a fixed pre-specified rule rather than per-task randomization. We did this because unconstrained randomization would

occasionally produce display orders where the three decoy options belonging to the same anchor would appear consecutively next to the anchor option itself. This is something we wanted to avoid, as such a clustering could have alerted attentive participants to the underlying construction logic of Part-2 tasks. The fixed ordering rule was designed to interleave anchor-based groups so that no such visually suspicious pattern would arise. Note also that the ordering rule was different between option sets.

The order of the 12 Part-1 tasks was randomized with one pre-defined constraint: positions 11 and 12 were occupied by two (randomly selected) tasks from the four Part-1 filler tasks that had no mapped counterpart in Part 2. Consequently, all eight mapped Part-1 tasks appeared within the first ten Part-1 positions, while still preserving randomization of task order subject to this constraint.

The procedure for each round of Part 2 was as follows: Participants saw the full table for five seconds. Then, they saw a different screen on which they had five seconds to choose between delegating to the algorithm corresponding to their treatment condition and choosing an option themselves. In the EV condition, delegation selected the lottery with the highest expected value. In the preference-based condition, delegation selected the non-dominated lottery which the participant had preferred in Part 1 over the other non-dominated lottery from the current task. Importantly, in both treatments the delegated option was always one of the two transformed anchor options and therefore never one of the six decoy options. Assignment to condition was randomized at the participant level. In the EV condition, participants were informed that "[the] algorithm works out how much each option would yield on average, and selects the option for which this is highest." In the preference-based condition, they instead read that "[the] algorithm analyses your choices from Part 1 and selects the best option for you in light of your Part-1 choices."

If they did not delegate, they saw the table containing the eight options for up to eight more seconds, during which they had to choose one of the options. After each Part-2 decision, feedback provided full resolution of uncertainty: participants were shown the chosen option and the realized payoff from one randomly drawn state of the world for that round, together with the payoffs that all eight options would have paid in that state. Each lottery had ten equally likely states, and the drawn states are consistent with this (chi-square test of uniformity across the ten states, $p = 0.23$). If they did not make a choice in the payoff-relevant task, they did not earn a bonus payment from the experiment. The total payoff from the experiment was GBP 2.00 plus the points earned in one randomly selected round of Part 2, converted at a rate of 1 penny per point (participants could earn up to GBP 8.90 in addition to the base payoff). On average, participants earned GBP 4.14 including the base payoff for a median time of just under 16 minutes.

3.3 Procedure

After being welcomed to the study, participants saw a symbolic representation of a bicycle (two red tires and two triangles in yellow and green that together formed the frame), being asked to indicate what that figure represented (a banana, a traffic light, darts games, a bicycle, a robot, or 'nothing, it's randomly placed shapes'). At the time, the agentic AIs would typically choose the traffic light, presumably because of the colouring of the shapes making up the bicycle. Following that, participants would read a broad overview of upcoming tasks ('In Parts 1 and 2: You participate in different decision-making tasks. In Part 3: You answer a short questionnaire', followed by the information that they would receive GBP 2 for participating and up to GBP 8.90 as a bonus, depending on decisions and on chance). Then, they would have to give consent to participating and to our using of their data for the purposes of this publication, which was followed by a final attention and 'system' check ('Please select "Disagree" on the following question' and 'please click on the white line in such a way that the appearing blue dot is close to a value of 50.').[4]

After these checks, Part 1 (preference elicitation) started, followed by a question on how satisfied they were with their choices from Part 1 overall. Then, participants completed Part 2 (delegation decisions with round-by-round outcome feedback), followed by a question of whether they would delegate to the algorithm in case they had to choose to delegate or not for 10 (hypothetical) additional tasks. After indicating that choice, participants estimated (on a 0–10 integer scale) in how many of those 10 tasks the algorithm would choose a non-preferred option and in how many it would choose a "completely weird" option. Before the final post-task questionnaire containing demographics and attitudinal items, we asked them for self-reported reasons for their choice in the hypothetical choice at the end of Part 2. Because our theoretical focus is reliance behavior, primary outcomes are behavioral rather than attitudinal.

3.4 Ethics approval

The study received ethics approval from the Institutional Review Board of the Gesellschaft für experimentelle Wirtschaftsforschung e.V. (GfeW; German Association for Experimental Economic Research), certificate no. 19HBJNwQ (status: approved; expedited review; approval date: 11/13/2025; certificate validity through 11/13/2027; see `https://gfew.de/ethik/19HBJNwQ`). All participants provided informed consent before participation.

[4] The 'system check' was meant to screen out certain types of bots, as, in an invisible box below the displayed white line, there was a '50' being displayed right next to the (visible) end-of-the-scale '100'.

3.5 Preregistered outcomes and analyses

The preregistered hypothesis (H1) predicted higher delegation to the EV-maximizing algorithm than to the preference-based algorithm. The preregistration document is available at `https://aspredicted.org/7un5qf.pdf`.

Primary preregistered analyses were:

1. a one-sided Boschloo exact test for the treatment difference in delegation in round 1 (first contact), and
2. a one-sided Wilcoxon–Mann–Whitney test comparing participant-level average delegation rates across Part 2.

We additionally estimated a preregistered linear probability model with decision-set fixed effects and participant-clustered standard errors, including treatment-by-period interactions and the share of outcomes with a payoff of zero among prior delegated choices. Formally, for participant i in decision-set (task) j at within-part decision number t, the preregistered specification is

$$\begin{aligned}\text{Delegated}_{ijt} = \beta_0 + \beta_1 \text{EV}_i + \beta_2 t + \beta_3 (\text{EV}_i \times t) \\ + \beta_4 \text{ZeroShare}_{i,<t} + \beta_5 (\text{EV}_i \times \text{ZeroShare}_{i,<t}) + \delta_j + \varepsilon_{ijt},\end{aligned} \tag{1}$$

where $\text{Delegated}_{ijt} \in \{0, 1\}$ is the binary delegation choice, EV_i is an indicator for the EV-algorithm condition, t indexes the within-part decision number (1–8), $\text{ZeroShare}_{i,<t}$ is the share of zero-payoff outcomes among participant i's prior *delegated* choices up to decision $t-1$ (defined as 0 for participants with no prior delegated choices), δ_j are decision-set (task) fixed effects, and standard errors are clustered at the participant level. We further estimated an exploratory specification excluding round 1 to separate first-contact effects from choices made after feedback experience. The preregistration additionally specified a secondary hypothesis on switching to non-delegation after a zero-payoff delegation; the hypothesis and our test of it are described in the subsection "Responses to feedback".

4 Results

4.1 Overview

Two empirical patterns organize the results. First, treatment differences are concentrated at first contact. Second, categorical non-delegation is substantially more common under preference-based personalization. Both patterns match the implications derived from the choice-authority framework in Section 1, and (as

we show in Section 4.6) they arise despite participants crediting the personalized algorithm with superior preference-matching, the mechanism signature that separates authority-based from accuracy-based accounts.

Table 1 summarizes the two preregistered primary tests, the preregistered secondary test on feedback, and the (non-preregistered) categorical non-delegation contrast, reported in full in the subsections below.

Table 1: Summary of the preregistered tests and the categorical non-delegation contrast.

Test	Outcome	Statistic	p-value	Effect size (two-sided 95% CI)
Boschloo exact (1-sided, prereg.)	Round-1 delegation	66.9% (EV) vs. 55.8% (Pref)	0.041	RD = 11.1pp (Wald [−1.3, 23.7]); OR = 1.60 ([0.91, 2.84])
Wilcoxon–Mann–Whitney (1-sided, prereg.)	Average delegation (rds. 1–8)	$W = 6428$	0.243	$r_{rb} \approx 0.05$ (boot. [−0.10, 0.20])
LPM interaction (1-sided, prereg. secondary)	Switch to non-delegation after a delegated round	$\beta_{\text{Pref}\times\text{zero payoff}}$	0.633	$\beta = -0.023$ (SE = 0.067)
Boschloo exact (1-sided, exploratory)	Categorical non-delegation	15.7% (Pref) vs. 3.4% (EV)	< 0.001	RD = 12.3pp (Wald [4.9, 19.7]); OR = 5.25 ([1.65, 22.07])

4.2 Primary test: first-contact delegation

As a benchmark for interpreting delegation behavior, choice quality among non-delegated decisions was low in both treatments. The share of non-dominated self-chosen options was 34.1% in the EV condition and 35.1% in the preference-based condition, implying that roughly two-thirds of self-made choices were dominated. Hence, participants had a clear instrumental reason to delegate. On average, delegated decisions yielded 243.49 points (GBP 2.43), compared to 195.18 points (GBP 1.95) for non-delegated decisions, a difference of approximately 25%.

In round 1, delegation was higher in the EV condition than in the preference-based condition (66.9% vs. 55.8%). The preregistered one-sided Boschloo exact test yields p = 0.041, supporting H1 for first-contact behavior. The corresponding effect size is a risk difference of 11.1 percentage points (Wald 95% CI $[-1.3, 23.7]$) and an odds ratio of 1.60 (Fisher's exact two-sided 95% CI [0.91, 2.84]). Note that the confidence intervals span 0 and 1, respectively, because we are reporting *two-sided* intervals here, while the pre-registered tests were one-sided tests. Figure 2 visualizes the round-level profile and shows that there is a clear treatment gap in the first decision.

4.3 Primary test: delegation across repeated decisions

At the participant level, the preregistered one-sided Wilcoxon–Mann–Whitney test on average delegation across Part 2 yields W = 6428 and p = 0.243. The corresponding effect size (rank-biserial correlation) is small, $r_{rb} \approx 0.05$ (bootstrap 95% CI $[-0.10, 0.20]$), consistent with a confidence interval that comfortably spans zero. We therefore do not find evidence of a treatment difference in average delegation across rounds.

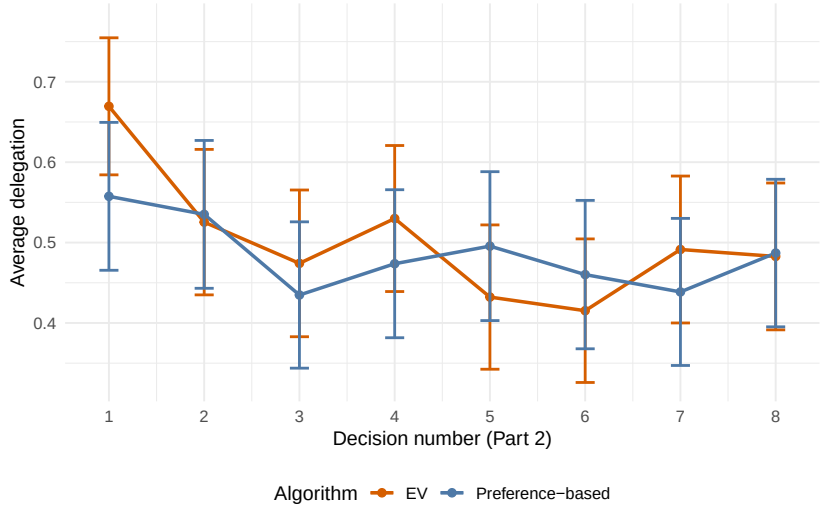


Figure 2: Average delegation over decisions in Part 2 by treatment condition. Whiskers indicate 95%-confidence bands.

Descriptively, this non-difference reflects post-first-contact convergence that is driven by a strong decline in delegation in the EV condition after round 1 (Figure 2). We treat this pattern as descriptive because our design was not optimized to identify why delegation declines more under EV after first contact. Notwithstanding, we will report on how this pattern relates to 'wins' (rounds in which the realized outcome was positive) and 'losses' (rounds in which the realized outcome was zero). We do not read the convergence as evidence that authority-based resistance to personalization is overridden by performance feedback. Two exploratory observations help to read it. First, average delegation in rounds 2–8 is the net of two offsetting differences of about six to seven percentage points each: more participants opt out entirely under personalization (Section 4.4), while delegation among participants who delegate at least once is lower under EV (49.8% vs. 56.4%). Second, part of the decline under EV is linked to the algorithm's picks. An EV pick that differs from the participant's own preferred option pays zero more often and is followed by less delegation (Section 4.5); had the 28 round-1 EV delegators with such a pick continued as often as those whose pick matched their preference, round-2 delegation under EV would have been about six percentage points higher (participant-level bootstrap 95% CI [0.4, 11.8]). By contrast, participants react to a zero payoff in the same way in both conditions, and EV picks produced only modestly more zero payoffs overall (40.3% vs. 36.2% of delegated decisions; participant-clustered logit $p = 0.18$). These decompositions are descriptive and imprecise (the difference between the two declines has $p = 0.14$ in a difference-in-differences logit on rounds 1 and 2), so much of the larger decline under EV

remains unexplained.

The preregistered regression specification points in a similar direction. The EVAlgorithm coefficient is small and statistically insignificant ($\beta = -0.049$, SE = 0.083, $p = 0.558$). Likewise, both interaction terms are imprecisely estimated (EVAlgorithm $\times$ WithinPartDecisionNumber: $\beta = 0.002$, SE = 0.012, $p = 0.871$; EVAlgorithm $\times$ share[Payoff $= 0$]: $\beta = -0.050$, SE = 0.098, $p = 0.612$), indicating no robust treatment difference over repeated decisions. By contrast, the share of prior delegated outcomes with a payoff of zero is strongly negatively associated with subsequent delegation ($\beta = -0.385$, SE = 0.078, $p < 0.001$).[5,6]

Taken together, the preregistered tests indicate that the treatment effect is concentrated at first contact and does not persist over repeated decisions, while negative feedback (in particular prior zero-outcomes) is strongly associated with lower subsequent delegation. The EV algorithm's picks carry a higher expected zero probability than the preference-based algorithm's (41.7% vs. 34.8% per delegated decision), but realized zero shares differ by only 4.1 percentage points (40.3% vs. 36.2%; $p = 0.18$). The null effect for rounds 2–8 reflects lower delegation under EV after round 1 (66.9% in round 1 vs. 48.1% in rounds 2–8; preference-based: 55.8% vs. 47.6%), not the preference-based algorithm gaining acceptance.

The regression evidence above is correlational across the whole sample, so we additionally probed the role of feedback using variation that is not confounded by the between-treatment difference in decision rule: conditional on having delegated in round 1, whether the realized payoff from that delegated choice happened to be zero or positive is plausibly as-good-as-random with respect to the participant's own dispositions, since it depends on which lottery state was drawn. Within the EV condition, round-1 delegators who received a zero payoff delegated again in round 2 only 36.0% of the time, versus 64.8% among those who received a positive payoff (two-sided Boschloo exact test $p = 0.020$; OR = 0.31, 95% CI [0.10, 0.91], based on Fisher's exact test). The contrast has the same sign and a similar size in the preference-based condition (50.0% vs. 72.5%; Boschloo $p = 0.078$; OR = 0.39, 95% CI [0.11, 1.29]). The two within-treatment contrasts do not differ significantly from each other (treatment-by-payoff interaction in a logistic regression: $p = 0.772$), so the weaker significance in the preference-based

[5] Running an analogous regression with fixed effects for the task on round-1 data only yields the same result as the Boschloo test, with $\beta = 0.113$, SE = 0.065, and $p = 0.041$ (one-sided).

[6] As a robustness check for the binary nature of the outcome, we re-estimated this specification as a logistic regression with the same fixed effects and cluster-robust standard errors. The qualitative pattern is unchanged: the EVAlgorithm coefficient remains small and insignificant ($b = -0.241$, OR = 0.79, $p = 0.549$), as do its interactions with period ($b = 0.014$, OR = 1.01, $p = 0.799$) and with the zero-outcome rate ($b = -0.241$, OR = 0.79, $p = 0.611$), while the zero-outcome history effect remains large, negative, and highly significant ($b = -1.661$, OR = 0.19, $p < 0.001$). The conclusions drawn from the linear probability model are therefore not an artifact of the linear functional form.

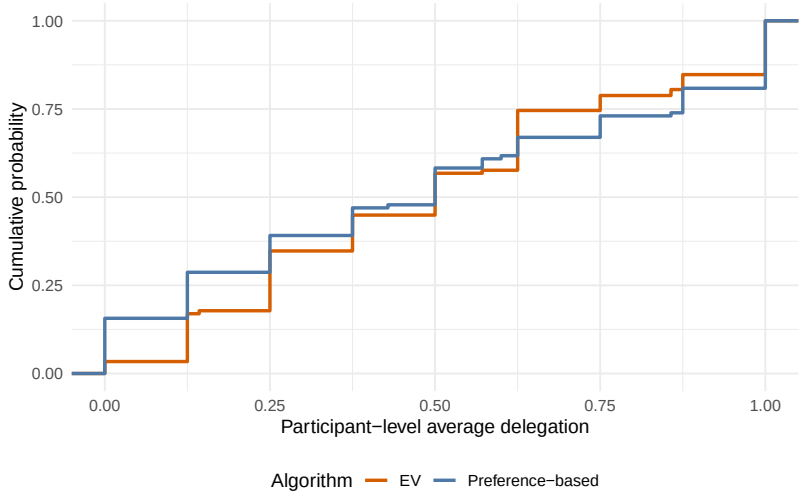


Figure 3: ECDF of participant-level average delegation by treatment condition.

condition mainly reflects the smaller number of round-1 delegators there (62 vs. 79).

The cross-participant distribution aggregated over all rounds is shown in Figure 3. Note that the convergence in delegation behavior over rounds does not diminish the substantive relevance of first-contact effects: in many real settings, the initial adoption decision can determine whether users return to a system at all. Theoretically, this time profile is also in line with the choice-authority account: rule construal is the dominant cue only while no outcome feedback is available; once feedback accumulates, experienced outcomes become an additional driver of delegation.

4.4 Categorical non-delegation

This section reports an outcome that was not preregistered (categorical non-delegation) that is complementary to the preregistered primary tests.

The second clear pattern is categorical non-delegation, which can be seen from the left end of Figure 3. In the preference-based condition, 18 of 115 participants never delegated (15.7%); in the EV condition, this applies to only 4 of 118 participants (3.4%). As this outcome was not preregistered, we treat the test as exploratory. A one-sided Boschloo exact test (in line with the directional prediction of the choice-authority framework that there will be more categorical non-delegation under personalization) yields $p < 0.001$ (two-sided: $p = 0.001$). The corresponding effect size is a risk difference of 12.3 percentage points (Wald 95% CI [4.9, 19.7]) and an odds ratio of 5.25 (95% CI [1.65, 22.07], based on Fisher's

exact test). This gap points to a distinct acceptance barrier under personalization for a subset of users.[7] Note that although there also seems to be a treatment difference in always-delegators, this difference is far less pronounced (19.1% vs. 13.6%). The subgroup structure of the effect is consistent with the heterogeneity implication of the choice-authority framework: if the intrinsic valuation of decision rights is concentrated in a subset of users (Bartling et al., 2014; Owens et al., 2014), personalization-induced resistance should manifest primarily as categorical opt-out rather than as a uniform downward shift in delegation.

Never-delegators were not meaningfully different in age from participants who delegated at least once (mean age 41.8 vs. 42.9 years; $t = -0.329$, $p = 0.745$), and we likewise find no clear association with household income (categorical; χ^2 = 13.171, df = 11, p = 0.282) or education (categorical; χ^2 = 1.859, df = 5, $p = 0.868$), and both spent similar amounts of time in the experiment (mean number of seconds 1113 for delegators vs. 1237 for never-delegators, $t = -0.971$, $p = 0.341$). And while there is a tendency for male participants to be never-delegators more often than females (12.0% vs. 6.3%), the evidence is weak (Boschloo exact test, $p = 0.156$). We also do not find meaningful differences in self-reported willingness to take financial risks (4.53 vs. 4.68; $t = -0.224$, $p = 0.825$), self-reported response reliability/care (6.91 vs. 6.91; $t = -0.059$, $p = 0.954$), or frequency of AI use/interactions (categorical: 5.87 vs. 5.18; $t = 0.866$, $p = 0.395$).[8] Within the preference-based condition, never-delegators actually reported slightly higher satisfaction with their Part-1 choices than delegators (7.94 vs. 7.56; $t = -1.400$, $p = 0.169$). This suggests that categorical non-delegation in this condition does not originate from dissatisfaction with the preference profile that served as the algorithm's input. These comparisons form an exploratory family of roughly ten tests that we did not correct for multiple testing, and the null results should be read with the sample size in mind: only 22 participants never delegated in total (18 in the preference-based condition, 4 in the EV condition), so the corresponding subgroup tests are underpowered to detect anything but large characteristic differences, and the absence of a significant contrast should not be read as evidence that never-delegators are indistinguishable from delegators on these dimensions.

The first-contact gap is concentrated among never-delegators. Among participants who delegated at least once, round-1 delegation is similar across conditions (preference-based: 65.6%, EV: 69.3%; two-sided Boschloo exact test $p = 0.64$).

[7] The same direction appears in an earlier study with a different design (reported in Suraci's PhD thesis): 32/190 never-delegators in preference-based treatments versus 17/187 in EV treatments (one-sided Boschloo exact test, $p = 0.015$).

[8] All t-tests reported in this subsection use the Welch correction (unequal variances), and all null conclusions are confirmed by the non-parametric Wilcoxon rank-sum equivalent (age: $p = 0.649$; time in experiment: $p = 0.105$; risk-taking: $p = 0.854$; reliability: $p = 0.956$; AI experience: $p = 0.445$).

Because never-delegation is defined over all eight rounds and therefore includes round 1, we repeat the comparison with "otherwise-never-delegators", that is, participants who never delegate in rounds 2–8. They make up 16.8% of the preference-based and 7.6% of the EV participants (one-sided Boschloo $p = 0.019$; two-sided $p = 0.037$), while round-1 delegation among all other participants is 64.9% and 67.9%, respectively ($p = 0.64$). The same holds when rounds 1 and 2 are analysed jointly and otherwise-never-delegators are defined over rounds 3–8 (23.2% vs. 12.7%; one-sided Boschloo $p = 0.022$). These comparisons are exploratory and descriptive, since they condition on later behavior.

4.5 Responses to feedback

We analyze how participants react to feedback in four steps that differ in status and in what they add. First, the preregistered secondary hypothesis asks whether a zero payoff from a delegated choice makes participants in the preference-based condition more likely than those in the EV condition to abandon delegation (confirmatory). Second, a win-stay/lose-shift model asks whether the two conditions react differently to any prior win or loss, using all rounds 2–8 rather than only rounds that follow a delegation (exploratory). Third, we ask whether a loss weighs more heavily when the algorithm rather than the participant produced it (exploratory). Fourth, we ask whether the EV algorithm's picks that differ from the participant's own preferred option are followed by less delegation (exploratory). The take-home message is that losses, and especially losses produced by the algorithm, push participants away from delegation in both conditions, with no evidence that this reaction is stronger under personalization; the EV algorithm's picks that differ from one's own preference are an additional, exploratory candidate for why delegation declines more under EV.

Preregistered test: switching to non-delegation after a zero payoff

In the preregistration, we suggested one secondary hypothesis on how participants respond to feedback: after delegating to the algorithm and obtaining a payoff of zero, participants in the preference-based condition are more likely to switch to non-delegation in the following round than under any other circumstances, in particular than under the same circumstances in the EV condition ("participants are less forgiving with respect to the PREF algorithm"). We did not pre-specify a test for this hypothesis. Here, we analyse it in a linear probability model of switching to non-delegation in those rounds 2–8 that follow a delegated round (799 observations from 209 participants), with task and decision-number fixed effects and participant-clustered standard errors. The regressors are an indicator for the preference-based condition, an indicator for a zero payoff in the

previous (delegated) round, and their interaction; the preregistered prediction corresponds to a positive interaction, which we test one-sided.

The data do not support our hypothesis. Participants did switch to non-delegation much more often after a zero payoff than after a positive payoff from a delegated choice, in both conditions (EV: 55.6% vs. 30.3%; preference-based: 45.3% vs. 21.5%). In the regression, this zero-payoff effect amounts to 0.254 (SE = 0.048) in the EV condition and 0.250 (SE = 0.048) in the preference-based condition ($p < 0.001$ in both). It is, however, not larger in the preference-based condition: the interaction is slightly negative and insignificant ($\beta = -0.023$, SE = 0.067, one-sided $p = 0.633$; two-sided $p = 0.734$). The conclusion does not depend on how "any other circumstances" is read. Restricting attention to rounds that follow a zero-payoff delegation, switching to non-delegation was, if anything, less frequent in the preference-based than in the EV condition ($\beta = -0.105$, SE = 0.070, two-sided $p = 0.140$). Comparing instead with all other rounds 2–8, including those that follow a self-chosen option or a positive payoff, yields an insignificant interaction of the wrong sign ($\beta = -0.071$, SE = 0.071, one-sided $p = 0.840$; 1,615 observations from 233 participants). Hence, participants were not less forgiving with respect to the preference-based algorithm.

Exploratory: win-stay/lose-shift

The remaining analyses in this subsection were not preregistered and are exploratory. To check whether the conclusion above depends on the narrow preregistered definition, we estimate a win-stay/lose-shift model on all rounds 2–8. It regresses an indicator for staying with the previous round's delegation/non-delegation choice on a prior win (a previous-round payoff larger than zero), the treatment, and their interaction, with task and decision-number fixed effects and participant-clustered standard errors. A prior win is strongly associated with staying (β_{lagWin} = 0.169, SE = 0.035, $p < 0.001$), and this association is statistically indistinguishable across treatments (EVAlgorithm $\times$ lagWin: β = 0.009, SE = 0.049, $p = 0.857$). In other words, participants in both treatments are more likely to switch after a loss than after a win, and the size of this reaction does not differ between treatments. The one treatment difference is a level shift: participants in the EV condition stay less often after a loss ($\beta_{\text{EVAlgorithm}} = -0.105$, SE = 0.050, $p = 0.036$), and, given the near-zero interaction, by a similar amount after a win. This reflects a general tendency of EV participants to switch more often rather than a stronger reaction to losses.

Exploratory: who produced the loss?

A more revealing picture emerges when we distinguish *who* produced the prior outcome, that is, whether the previous round was delegated (so that "the algorithm lost or won") or self-chosen. Estimating the model separately by treatment and interacting a prior loss with prior delegation, the prior-loss × prior-delegation interaction is negative and significant in both conditions (EV: $\beta = -0.185$, SE = 0.070, $p = 0.009$; preference-based: $\beta = -0.140$, SE = 0.064, $p = 0.030$). That is, a loss prompts more switching in general, but a loss that the algorithm produced prompts substantially more switching than a self-inflicted loss. This is visible in the raw stay rates: after a delegated loss, the stay rate is the lowest in both conditions (54.7% in the preference-based and 44.4% in the EV condition), well below the stay rate after a delegated win (78.5% and 69.7%) and even below that after a self-chosen loss (67.7% and 58.3%). Note that "staying" means continued delegation after a delegated outcome but continued self-choice after a self-chosen one, so these comparisons are descriptive. This exploratory pattern connects to the broader finding that algorithmic mistakes tend to be forgiven less readily than one's own (Dietvorst et al., 2018; Renier et al., 2021): participants abandon delegation disproportionately after the algorithm delivers a bad outcome, even though delegated choices were on average substantially more profitable. The preregistered prediction that this reaction would be stronger toward the preference-based than toward the EV algorithm was, however, not supported. The reaction to a given loss is therefore similar across treatments, which reinforces the message that the treatment difference is concentrated at first contact, rather than in how participants respond to a loss thereafter.

Exploratory: when the algorithm's pick differs from one's own preference

Under the preference-based algorithm, the delegated option is by construction the participant's own preferred option. Under EV, it coincides with that option in about two thirds of delegated rounds. Table 2 splits the delegated rounds 1–7 accordingly. EV picks that differ from the participant's preference pay zero more often (50.8%, against 36.3% for matching EV picks and 36.0% under preference-based delegation) and are followed by less delegation, especially after a zero payoff. Delegation after matching EV picks is not significantly different from delegation under preference-based delegation (logit coefficient -0.38, participant-clustered $p = 0.11$; zero shares are virtually identical). Mismatching EV picks stand out (-0.63 relative to preference-based delegation, $p = 0.021$; -0.74, $p = 0.010$ with task fixed effects). Controlling for a zero payoff reduces this only partly (-0.50, $p = 0.072$; -0.67, $p = 0.023$ with task fixed effects). Within the EV condition, a zero payoff weighs somewhat more heavily after a mismatching pick (interaction -0.73, $p = 0.093$). The mismatch is, however, not randomly assigned, since it depends on the participant's own preferences, and the pooled EV coefficient falls

only from -0.46 to -0.43 when a zero payoff is controlled for. We therefore read these results as pointing to one contributing channel, not as a full account of why delegation declines more under EV. The forgone payoffs that the feedback screen showed (the best available option minus the realized payoff) predict continued delegation on their own but not once a zero payoff is controlled for (see the replication code).

Table 2: Delegated rounds 1–7 by whether the algorithm's pick matched the participant's own Part-1 preference (exploratory).

			Delegates again in the next round		
Group	n	Zero payoff	Overall	After a zero	After a positive payoff
Preference-based	386	36.0%	69.9%	54.7%	78.5%
EV, pick matches preference	281	36.3%	61.2%	49.0%	68.2%
EV, pick differs from preference	132	50.8%	55.3%	37.3%	73.8%

4.6 Exploratory mechanisms: expectations and post-task reasons

To better understand the behavioral patterns above, we draw on two sets of exploratory measures collected after Part 2: participants' quantitative expectations about how the algorithm would perform, and their self-reported reasons for a hypothetical further delegation choice. We organize the discussion around three questions: (i) Did participants believe the preference-based algorithm actually captured their preferences? (ii) Which expectations are most tightly linked to delegation, and does this differ by algorithm type? (iii) What do the open- and closed-item reasons reveal about the motives of those who delegate sometimes or always versus those who never do?

Did participants trust the personalized algorithm to match their preferences?

A natural concern about our design is that lower delegation under personalization might simply reflect participants doubting that the preference-based algorithm captured their preferences in the first place. The expectation data speak directly against this concern. After the incentivized rounds, participants estimated, out of 10 hypothetical further tasks, in how many tasks the algorithm would pick a non-preferred option and in how many it would pick a "completely weird" option. Participants in the preference-based condition expected the algorithm to make markedly *fewer* non-preferred choices than those in the EV condition

(on average 3.78 vs. 5.00 out of 10; $t = -4.15$, $p < 0.001$; Wilcoxon rank-sum $p < 0.001$). Expectations about "completely weird" choices, in contrast, did not differ across conditions (1.57 vs. 1.78; $t = -0.94$, $p = 0.351$). Participants thus clearly credited the personalized algorithm with doing its job: relative to objective optimization, it was expected to select options aligned with their preferences substantially more often, without any rise in the perceived risk of grossly erratic choices. This holds even for the participants who opted out entirely: within the preference-based condition, never-delegators expected just as few non-preferred choices as delegators (3.83 vs. 3.77 out of 10; $t = -0.11$, $p = 0.912$). Categorical non-delegation under personalization thus cannot be attributed to a belief that the algorithm failed to capture one's preferences, reinforcing the earlier finding that these never-delegators were, if anything, slightly *more* satisfied with the Part-1 choices that fed the algorithm. Note further that if we restrict the sample to only those participants who expect a maximum of 2 mistakes by the algorithm (21% of the participants), our treatment differences become stronger rather than weaker, adding further evidence that the treatment effects do not seem to be driven by participants not believing in the ability of the preference-based algorithm to do its job.

Which expectations are most tightly linked to delegation?

This makes the reduced delegation under personalization all the more striking. Even though participants believed the preference-based algorithm would more often choose in line with their preferences, they were no more (and at first contact significantly less) willing to delegate to it. The barrier therefore does not appear to lie in doubts about the algorithm's accuracy. Consistent with this, the two explanation-demand items ("I find it unpleasant that I don't know how exactly the computer chooses" and "I would have preferred more information about how the computer makes its decisions") were selected only infrequently, as was the profiling-related item ("I don't like to be analysed by an algorithm"). Participants' reservations were thus not primarily about understanding, transparency, or competence at preference-matching, pointing instead toward concerns about ceding choice authority itself, as we will show below. This is precisely the mechanism signature derived in Section 1: higher perceived preference-fit combined with equal or lower delegation speaks against accuracy-based trust accounts and is consistent with authority-based resistance (although, as noted in the Limitations, the mechanism is not causally identified).

What distinguishes delegators from never-delegators?

The closed-item reasons (Figure 4) vary more by delegation type than by treatment. Note that the percentages discussed in this subsection pool delegators and never-delegators across both treatments, whereas Figure 4 itself displays the same items split by treatment condition; the two views are complementary but not numerically identical, and the pooled figures reported here cannot be read directly off the by-treatment bars. The only visible treatment contrast—larger never-delegator/delegator gaps in the EV condition—we do not interpret, because the EV never-delegator subgroup comprises only four participants. Two patterns stand out. First, participants who delegated at least once were far more likely to cite performance-based motives: 70.1% indicated that the computer would choose better, against only 36.4% of never-delegators (that they would choose better). Second, never-delegators more often selected reasons centered on general distrust of the algorithm (50.0% vs. 32.2%; recall, however, that those who indicate the strongest belief in that the algorithm does its job do not exhibit a smaller treatment effect; if at all, the effect is stronger in that subset) and on retaining personal responsibility for the choice (63.6% vs. 38.9%). At first glance, the prevalence of the "distrust" item among never-delegators might seem to sit uneasily with our claim that resistance is not accuracy-based: the expectation data (above) show that participants, including never-delegators, expected the personalized algorithm to match their preferences at least as well as the objective rule, yet about half of never-delegators still endorsed a distrust-worded item. We think this reflects a conflation built into the closed-item wording itself rather than a contradiction in the evidence: the item ("I don't fully trust the computer") does not distinguish *distrust of the algorithm's accuracy* from a more diffuse *distrust of ceding the choice to any external party*, including one the respondent otherwise believes to be competent. Our quantitative expectation measures speak specifically to the former and speak against it as the dominant driver; the qualitative reason item plausibly picks up a mixture in which the latter, authority-related sense of "distrust" predominates among never-delegators. Future work using closed items that separate these two senses of distrust explicitly would allow a cleaner test of this reading. This mirrors the treatment-level non-delegation pattern and is consistent with a subgroup that weights agency and responsibility concerns more heavily than expected performance. Among the remaining items, finding the task too tedious appears (at a low level) only among delegators (10.9% vs. 0%), consistent with an effort-saving channel, and open-text responses repeatedly mention time pressure, a general choice-friction channel that does not isolate a treatment-specific mechanism.

These mechanism patterns should be interpreted cautiously. The reason measures are ex-post and refer to a hypothetical additional choice rather than to

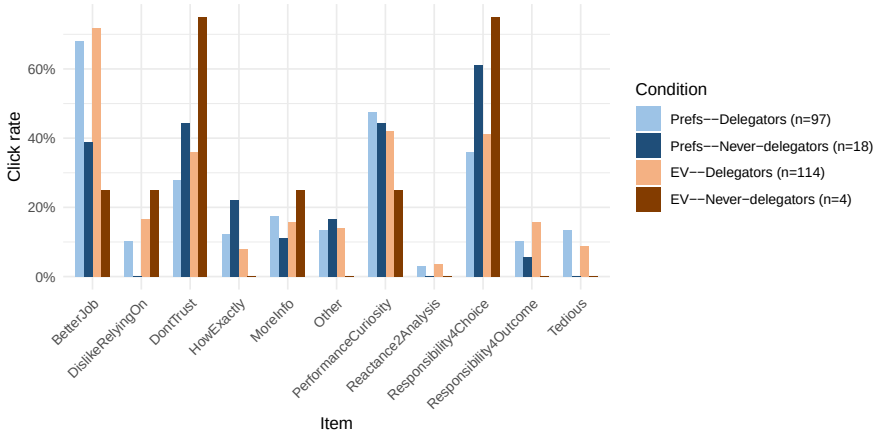


Figure 4: Distribution of reasons indicated for delegating or not 10 additional hypothetical tasks to the algorithm. Item labels correspond to the following verbatim statements presented to participants (wording adapted to whether the participant had chosen to delegate or not): *DontTrust* – "I don't fully trust myself/the computer"; *BetterJob* – "I thought the computer/I could do a better job"; *HowExactly* – "I find it unpleasant that I don't know how exactly the computer chooses an option for me"; *MoreInfo* – "I would have preferred more information about how the computer makes its decisions"; *Responsibility4Choice* – "I didn't want/wanted to be responsible for choosing an/the option"; *Responsibility4Outcome* – "I didn't want to be responsible if a low payment results"; *DislikeRelyingOn* – "I don't like relying on myself/computers"; *Reactance2Analysis* – "I don't like to be analysed by an algorithm"; *PerformanceCuriosity* – "I wanted to see how well the computer/I can do"; *Tedious* – "Doing it myself would have been too tedious" (delegators only).

the incentivized rounds themselves, so they function as suggestive proxies for motives rather than as causal identification of psychological channels.

5 Discussion

5.1 Core finding

This preregistered study shows that preference-based personalization can deter delegation rather than increase it. We observe this deterrence in two forms: lower delegation at first contact and a larger subgroup of participants who never delegate under personalization.

The first-contact finding is important because it captures behavior before

participants can learn from outcome feedback. At that stage, delegation is driven primarily by how participants interpret the algorithmic rule itself. Our evidence is consistent with the interpretation that an impersonal EV rule is initially easier to accept as a legitimate external computation device than a profile-based personalized rule based on the user's own revealed preferences. Even though subsequent rounds show convergence in average delegation (driven by reduced delegation to the EV algorithm), first-contact differences remain practically important because initial adoption decisions can have path-dependent downstream effects in real-world deployment where customers may not constantly be prompted to make a delegation choice like in our experiment (think of immediate opt-out, lower willingness to return, and weaker opportunity for learning-by-use).

5.2 Theoretical implications

Our findings advance the theoretical understanding of algorithm adoption in three ways. First, they show that algorithm aversion is not a unitary phenomenon. The literature has largely treated reluctance to rely on algorithms as a problem of miscalibrated trust in algorithmic *competence*, to be remedied by demonstrating accuracy, providing explanations, or allowing output adjustment (Dietvorst et al., 2018; Leichtmann et al., 2023; Yeomans et al., 2019). Our evidence is most consistent with a second, conceptually distinct channel: authority-based resistance. Participants credited the personalized algorithm with *superior* preference-matching, and never-delegators were, if anything, more satisfied with the preference input that fed the algorithm—yet delegation was lower at first contact and categorical opt-out was more than four times as prevalent. Interventions targeting accuracy beliefs would not be expected to move this margin; indeed, among participants with the strongest confidence in the algorithm's competence, the treatment effect was stronger, not weaker. Competence-oriented and authority-oriented barriers therefore call for different remedies, and conflating them risks systematically misdiagnosing adoption failures.

Second, the results identify *rule construal* as a theoretically meaningful determinant of delegation, distinct from the interface, control, and communication features studied so far (Dietvorst et al., 2018; Fink et al., 2024; Hildebrand & Bergner, 2021). Holding the delegation architecture fixed, merely changing what the algorithm is said to optimize (an impersonal criterion versus the user's own preferences) changed behavior. In our perspective, this is because the decision rule shapes whether delegation is construed as outsourcing a computation or as surrendering one's judgment, and only the latter construal affects the intrinsic value people attach to holding decision rights (Bartling et al., 2014; Frey et al., 2004; Owens et al., 2014). This links the algorithm-adoption literature to a well-established body of evidence on choice authority and autonomy (Brehm,

1966; Deci & Ryan, 2000; Leotti et al., 2010) that has so far played little role in explaining human–algorithm interaction, and it generates testable predictions: any design cue that strengthens the judgment-substitution construal (e.g., emphasizing that the system "knows you" or "chooses as you would") should increase resistance among high-authority-valuation users, while cues that support the computation-outsourcing construal should reduce it.

Third, the heterogeneity structure of the effect, namely the concentration in categorical opt-out rather than a uniform shift of delegation propensities, matters for how personalization effects should be theorized and measured. If the intrinsic valuation of decision rights is heterogeneous (Bartling et al., 2014; Owens et al., 2014), average treatment effects on delegation understate the behavioral consequences of personalization for the affected subgroup, who exit entirely. This has a general methodological implication for research on technology acceptance: distributional and subgroup analyses (such as our cumulative density functions and never-delegator comparisons) can reveal adoption barriers that mean comparisons conceal.

Finally, our two-stage pattern suggests a simple integrative account of when rule characteristics matter: construal of the decision rule dominates at adoption, when no experiential evidence exists, whereas outcome feedback and the well-documented asymmetric forgiveness of algorithmic errors (Dietvorst et al., 2015; Renier et al., 2021) appear to dominate during continued use: in our exploratory feedback analyses, participants withdrew from delegation disproportionately after a bad outcome in both conditions, but we do not find that differential feedback explains the convergence. Personalization thus operates on the adoption margin, while error tolerance operates on the retention margin; theories of algorithm acceptance should model these margins separately.

5.3 Interpretation and relation to prior work

The categorical non-delegation pattern is particularly informative: many participants entirely avoid delegation under personalization, consistent with work on autonomy, responsibility, and control as drivers of acceptance. Exploratory post-task reasons (Section 4.6) align with this interpretation, especially the stronger emphasis among never-delegators on responsibility and distrust, while concerns about being algorithmically analyzed appear comparatively rare. Because these reason measures are ex-post and hypothetical, this evidence remains suggestive rather than causal. Crucially, the lower delegation under personalization does not stem from participants doubting that the personalized algorithm captured their preferences: they expected it to make significantly fewer non-preferred choices than the EV algorithm (3.78 vs. 5.00 out of 10; $p < 0.001$), while expectations about "completely weird" choices did not differ across conditions. Participants

thus credited the preference-based algorithm with successfully matching their preferences, yet were no more willing—and at first contact less willing—to delegate to it. This strengthens the interpretation that the relevant barrier is the act of ceding choice authority itself rather than skepticism about the algorithm's accuracy. A complementary channel operates through how algorithmic errors are treated once they occur: in both conditions, participants withdrew from delegation disproportionately after the algorithm produced a bad outcome (an exploratory finding; Section 4), echoing evidence that algorithmic mistakes are forgiven less readily than one's own (Dietvorst et al., 2018; Renier et al., 2021). The preregistered prediction that participants would be less forgiving of the preference-based than of the EV algorithm was not supported.

Two further accounts could produce similar behavior and deserve mention. A responsibility-offloading account holds that people delegate to an objective rule because it gives them someone to blame for bad outcomes, whereas a personalized rule leaves them feeling responsible for its errors. Our data speak to this only partly. Never-delegators endorsed the item on not wanting to be responsible for the choice far more often than delegators (63.6% vs. 38.9%), which fits retaining responsibility better than offloading it, and an HR experiment found that blame avoidance was a weak motive for delegation relative to algorithm aversion (Maasland & Weißmüller, 2022). Closer to our setting, participants who had to delegate a task affecting a third party chose a machine over a human only about a quarter of the time (26.5%), and neither lower perceived utility, lower trust, nor a weaker ability to pass on responsibility to a machine accounted for this (Gogoll & Uhl, 2018). Because everyone had to delegate to someone in that design, it speaks to the choice between agents rather than to the choice between delegating and deciding oneself, but it likewise suggests that a pure blame-shifting motive does not explain reluctance to rely on algorithms. We did not measure anticipated blame directly, so offloading motives among delegators cannot be excluded. A psychological-ownership account is also compatible with ours: if a system decides as the user would, the resulting choice may feel less like the user's own. Work on AI-generated text finds that users do not feel they own what the AI produced and that their sense of ownership rises with their own influence on the output (Draxler et al., 2024). This is conceptually close to our choice-authority account. That evidence concerns text generation rather than delegated choices, though, and personalization did not alter ownership there, so it does not directly predict our result. Ownership and anticipated-blame measures are natural additions for future work.

These findings are consistent with an earlier study on the same question (reported as a chapter in Suraci's PhD thesis), which motivated and informed the design of the present preregistered experiment. In that study, the delegation gap between EV and preference-based treatments was larger (about 20 percentage

points) and persisted over decisions. Recomputing the categorical non-delegation contrast for that earlier sample also yields a significant difference (32/190 in preference-based treatments vs. 17/187 in EV treatments; one-sided Boschloo exact test, $p = 0.015$).[9]

5.4 Limitations

The findings should be interpreted in light of several limitations. First, the study is an online experiment with stylized financial lotteries, so external validity to field settings should be interpreted cautiously. The Part-2 lottery choices were real and incentivized, not hypothetical vignettes: one round was paid out for real money, contingent on the participant's own choice or, if delegated, on the algorithm's choice, so the delegation decisions we analyze carried genuine financial consequences rather than being merely described. What remains stylized is the task *domain*: our lotteries abstract away features of real investment or medical decisions such as ambiguous or delayed payoffs, product complexity, and social or institutional framing (e.g., a named bank or physician), any of which could independently affect how the loss of choice authority is experienced. This abstraction is what let us isolate the construal mechanism cleanly (see the theoretical framework above), but it also means we cannot yet speak to whether the same first-contact resistance and categorical opt-out would emerge, and at what magnitude, once real-world complexity and institutional trust cues are reintroduced; we return to this as a direction for future work below. Second, although behavioral outcomes are well suited for studying reliance, the design does not causally identify the psychological mechanism behind first-contact hesitation or categorical non-delegation (e.g., perceived objectivity, profiling concerns, or accountability salience); our expectation data speak against accuracy-based accounts, and the post-task reasons point toward choice authority, but a direct manipulation of authority salience remains for future work. Relatedly, the two algorithm descriptions differ not only in decision rule but also in linguistic register: the EV description is phrased mechanically ("works out how much each option would yield on average"), while the preference-based description is phrased more personally ("analyses your choices…and selects the best option for you"). We cannot fully rule out that some of the effect reflects this difference in tone rather than construal of the rule per se. We note, however, that our earlier study using more symmetric wording for the two algorithms ("uses mathematics to work out…" versus "uses modern decision science and their choices…to model their behaviour") found an even larger delegation gap (about 20 percentage points,

[9]Note that categorical non-delegation was not analyzed or reported in the earlier study, which is the reason for why we did not pre-register its analysis for the present study.

versus 11.1 here), suggesting that the effect does not depend on the specific asymmetry in the present wording. A cleaner test would equate the two descriptions' register explicitly in future work. Third, the EV and preference-based treatments differ along several dimensions at once: personalization narrowly defined, the perceived objectivity of the decision criterion, whether the rule explicitly draws on the user's own prior choices (profiling), the criterion being optimized, and possibly expectations about financial performance. Our design therefore identifies a difference between two decision rules rather than a pure personalization effect. The data help to narrow down the candidates, but they do not eliminate them. Participants in the preference-based condition expected the algorithm to match their preferences better, not worse, which speaks against a lower-competence account at the level of preference matching. We did not, however, elicit expectations about payoffs, and the EV description promises the highest average yield, so a payoff-expectation account cannot be fully excluded. A profiling-aversion account is unlikely to be the dominant driver, because the profiling-related reason item ("I don't like to be analysed by an algorithm") was selected only infrequently. What remains is the combination of perceived objectivity and the preference-based rule's explicit reference to the user's own judgment, which is the dimension our choice-authority account targets. The present design cannot separate the two, and future work could manipulate them independently (e.g., an objective rule that is explicitly based on the user's own revealed data). Relatedly, outcome feedback is generated by the rule a participant is assigned to, so it can differ across conditions. In our data the difference in exposure to zero payoffs is small (40.3% vs. 36.2% of delegated decisions), but we cannot exclude that feedback contributes to the convergence in average delegation (Figure 2). Our design also identifies the difference between the two rules, not whether the objective rule raises or the personalized rule lowers delegation relative to a neutral baseline. This caveat does not extend to our two central results: the first-contact difference is measured before any outcome feedback exists, and categorical non-delegation by definition involves participants who never received algorithm-generated feedback. Fourth, low choice quality in the self-chosen decisions (roughly two-thirds of self-made choices were dominated, see "Primary test: first-contact delegation" below) raises the question of whether some non-delegation reflects task difficulty rather than a genuine valuation of choice authority. We consider this unlikely to explain categorical non-delegation specifically: never-delegators' own non-dominated choice rate among their self-made decisions was statistically indistinguishable from that of participants who delegated at least once (participant-level averages of 34.7% for the 22 never-delegators vs. 33.6% for the 170 participants who delegated at least once but also made at least one decision themselves; Welch $t = -0.25$, $p = 0.806$; Wilcoxon $p = 0.669$), so never-delegators were not making better- or worse-informed choices than delegators, and if anything, participants who found the

task difficult would be expected to delegate more, not less, which is the opposite of what categorical non-delegation implies. We cannot rule out that task difficulty contributes to delegation decisions among participants who delegate only sometimes, which future work could probe directly. Fifth, the present analyses focus on short-run repeated interaction (eight incentivized rounds within a single session); longer-run adaptation to personalized systems remains an open question. Our within-session data already show that the treatment gap in average delegation narrows after the first round, which is at least consistent with some short-run adaptation as participants gain outcome feedback. However, this convergence occurred over a handful of rounds in one sitting and should not be extrapolated to long-term human–computer interaction in real investment scenarios, where the relevant time scale is months or years rather than minutes. Extended real-world use plausibly triggers mechanisms absent from our design in both directions: on one hand, an accumulating track record of satisfactory outcomes and growing familiarity with a personalized advisor could erode authority-based resistance and allow accuracy-based trust to dominate, consistent with evidence that reliance builds gradually through experience; on the other hand, real investment relationships involve continued or deepening profiling (the system knowing progressively more about the user), higher stakes, and switching costs that could instead sustain or amplify the loss of choice authority we document. Because our design cannot adjudicate between these possibilities, we see longitudinal field studies that track delegation to personalized versus objective financial decision aids over repeated, consequential interactions as a necessary next step, and we flag this explicitly as a direction for future research below rather than assume our first-contact and short-run results generalize to long-term use. Sixth, our preregistration set the sample-size target (120 participants per condition, no fewer than 100 after exclusions) without stating an explicit power analysis. We reconstruct here what such an analysis would have shown using only the information available before the present study was run: the effect sizes from our earlier study reported in Suraci's PhD thesis. In that study, the EV-versus-preference-based gap was approximately 22 percentage points at first contact (67.4% vs. 45.8%, pooling across an additional feedback-timing factor absent from the present design) and approximately 19 percentage points in average delegation across all decisions. At $n = 100$–120 per arm, the minimum detectable risk difference at 80% power (one-sided $\alpha = 0.05$) for a first-contact-type proportion comparison is approximately 16–17 percentage points, comfortably below the earlier study's first-contact gap; correspondingly, power to re-detect an effect of the earlier study's magnitude was approximately 91–95% (exact-test simulation; 93–96% under the standard normal approximation). For the average-delegation (Wilcoxon–Mann–Whitney-type) comparison, the analogous minimum detectable effect is approximately 10–11 percentage points (using the earlier study's within-condition variance), again well

below the earlier study's roughly 19-percentage-point gap, with approximately 99.7–99.9% power to re-detect an effect of that size. On this basis, a sample of 100–120 per condition would have looked more than adequately powered at the design stage. With hindsight, the first-contact gap we actually observed in the present, more tightly controlled design was smaller than the earlier study's gap (11.1 percentage points, 66.9% vs. 55.8%, $p = 0.041$ one-sided), falling below the design's 16–17-percentage-point detection threshold, which offers a natural explanation for why this confirmed effect is only marginally significant. We deliberately stop short of computing a "power" figure for this realized effect itself: retrospective power calculated from a study's own observed effect size is a deterministic function of the p-value and provides no additional information beyond it (Hoenig & Heisey, 2001). For the same reason, we do not interpret the null average-delegation result ($p = 0.24$) through a post-hoc power lens; instead, the regression evidence and the round-level pattern in Figure 2 indicate that the EV-versus-preference-based difference genuinely converges to approximately zero after round 1, mainly through a decline in the EV condition (see "Primary test: delegation across repeated decisions" above). Seventh, our sample was recruited through Prolific from a UK-based participant pool; while we see no principled reason to expect the direction of the effect to be culture- or platform-specific, the magnitude of resistance to ceding choice authority could plausibly vary with cultural background, familiarity with algorithmic systems, or the demographic composition of online panels, and we have not tested this.

5.5 Implications and future research

For designing algorithmic decision systems, the results imply that personalization should not be treated as a uniformly beneficial acceptance lever. In settings where objective criteria are salient and normatively legitimate, objective optimization may face lower initial resistance than preference-based personalization. This has direct implications for the design and communication of decision aids in finance and other high-stakes domains.

These extensions are speculative given the abstract, single-domain, and time-constrained nature of our lottery task, and we offer them as hypotheses for domain-specific replication rather than as established implications. Financial robo-advisors routinely face the choice between objective portfolio rules (e.g., mean-variance optimization) and preference-based personalization derived from risk profiling questionnaires (Capponi et al., 2022; Rühr, 2020). Our findings suggest that the very feature intended to increase subjective fit—tailoring the algorithm to the user's profile—may inadvertently encroach on users' sense of choice authority, deterring precisely those users who value exercising active control over their decisions. The same tension arises whenever algorithmic decision aids must be

voluntarily adopted: in healthcare, treatment recommender systems can follow clinical-evidence guidelines or patient-preference protocols; in hiring, screening tools can apply standardized criteria or adaptively weight candidate profiles. In each case, the assumption that preference alignment promotes adoption deserves empirical scrutiny rather than being taken as a design axiom.

Future research should test mechanism-targeted interventions at first contact, including reframing of personalization logic, responsibility framing, and alternative explanation formats. The construal account developed here suggests concrete candidates: presenting a personalized system as a tool that *implements* the user's prior decisions (computation-outsourcing framing) rather than one that *decides as the user would* (judgment-substitution framing) should reduce authority-based resistance without changing the underlying rule. Measuring individuals' intrinsic valuation of decision rights (Bartling et al., 2014; Owens et al., 2014) before exposure would additionally allow a direct test of the predicted heterogeneity. It should also examine whether the non-delegator subgroup is stable across domains, stakes, and cultural contexts.

5.6 Data and code availability

Data, analysis code, and study materials will be deposited in a public repository upon acceptance/publication of this manuscript. In the meantime, a replication package containing the de-identified participant-level and decision-level data, the full analysis script reproducing every statistic and figure reported in this paper, and the complete set of Part-1 and Part-2 payoff matrices (including the display-order rule referenced in Section 3) has been made available to the editor and reviewers for the purposes of this review. We have opted not to make this package publicly accessible prior to acceptance.

5.7 Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude Opus 4.8 in order to assist in content organization as well as improving language and readability of both the manuscript and the analysis code (at the same time as allowing for an additional check on the statistical analyses). After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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A Appendix: Full task matrices

This appendix reports the complete numeric payoff matrices for all 12 Part-1 binary lottery-choice tasks and all 8 Part-2 option sets actually used in the experiment, as referenced in Section 3. Columns S1–S10 give the payoff (in points) in each of the ten equiprobable states; "Order" (Table 3) gives an example of the randomized presentation order of the 12 Part-1 tasks; "P1 task" and "P2 order" (Table 4) link each Part-2 option set back to its Part-1 anchor task and give its randomized position within Part 2 (for the particular realization of the randomization displayed). In Table 4, the eight options listed for each OptionsetID are the transformed anchor options (derived from the corresponding Part-1 task) plus the six state-wise dominated decoy options described in Section 3; they are reproduced here in the exact within-task display order used on screen.

Table 3: Part-1 lottery tasks (12 binary choices). Each row is one option; "Opt" links each option to the corresponding Part-2 option in Table 4. The variable "Ord." gives an example for the randomized presentation orders used in the experiment.

OSID	Opt	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	Ord.
1	3	525	525	525	525	525	0	0	0	0	0	2
1	6	225	225	225	225	225	225	225	205	0	0	2
2	5	670	670	670	610	0	0	0	0	0	0	3
2	7	255	255	255	255	255	235	205	0	0	0	3
3	2	625	625	625	625	525	0	0	0	0	0	7
3	7	290	290	290	290	290	290	230	0	0	0	7
4	4	390	390	390	390	360	330	0	0	0	0	6
4	8	890	890	860	800	0	0	0	0	0	0	6
5	4	340	340	340	340	340	0	0	0	0	0	9
5	8	235	235	235	235	235	235	235	205	0	0	9
6	4	435	435	435	405	0	0	0	0	0	0	1
6	5	265	265	265	265	265	265	235	0	0	0	1
7	4	320	320	320	320	320	320	220	0	0	0	5
7	5	525	525	525	505	0	0	0	0	0	0	5
8	4	390	390	390	390	390	360	330	0	0	0	4
8	8	860	860	830	0	0	0	0	0	0	0	4
9	6	525	525	525	505	0	0	0	0	0	0	10
9	7	230	230	230	230	230	230	230	200	0	0	10
10	6	260	260	260	260	260	260	230	0	0	0	12

OSID	Opt	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	Ord.
10	7	435	435	435	435	405	0	0	0	0	0	12
11	2	860	860	830	0	0	0	0	0	0	0	11
11	5	280	280	280	280	280	280	260	230	0	0	11
12	4	585	585	585	585	535	0	0	0	0	0	8
12	7	390	390	390	390	390	360	0	0	0	0	8

Table 4: Part-2 lottery tasks (8 option sets of 8 options each, as used in the experiment). The presentation within each Task/Option set ("OSID") followed the displayed fixed pre-specified order in the Table (by variable "Opt"); P1 links each set back to its Part-1 anchor task, P2 gives its randomized position in Part 2.

OSID	Opt	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	P1	P2
1	1	225	225	0	225	0	225	225	225	225	0	2	4
1	2	525	0	505	0	525	0	0	0	525	505	2	4
1	3	525	0	525	0	525	0	0	0	525	525	2	4
1	4	225	225	0	225	205	205	205	225	225	0	2	4
1	5	225	205	0	205	205	225	225	205	225	0	2	4
1	6	225	225	0	225	205	225	225	225	225	0	2	4
1	7	525	0	525	0	525	0	0	0	225	525	2	4
1	8	505	0	525	0	505	0	0	0	505	525	2	4
2	1	0	0	670	0	0	610	670	0	0	610	3	1
2	2	0	255	235	235	205	0	235	255	235	0	3	1
2	3	0	255	205	255	205	0	235	255	255	0	3	1
2	4	0	235	255	255	205	0	235	235	255	0	3	1
2	5	0	0	670	0	0	610	670	0	0	670	3	1
2	6	0	0	370	0	0	610	670	0	0	670	3	1
2	7	0	255	255	255	205	0	235	255	255	0	3	1
2	8	0	0	610	0	0	610	610	0	0	670	3	1
3	1	0	0	625	0	525	0	625	0	625	525	7	7
3	2	0	0	625	0	625	0	625	0	625	525	7	7
3	3	290	290	0	0	230	290	230	290	290	0	7	7
3	4	0	0	525	0	625	0	625	0	625	525	7	7
3	5	260	260	0	0	290	290	230	290	290	0	7	7
3	6	290	290	0	0	290	290	230	290	200	0	7	7
3	7	290	290	0	0	290	290	230	290	290	0	7	7
3	8	0	0	605	0	625	0	625	0	605	525	7	7
4	1	360	390	0	360	0	390	0	0	360	330	6	6
4	2	360	390	0	390	0	300	0	0	390	330	6	6

OSID	Opt	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	P1	P2
4	3	890	0	800	0	0	0	800	0	0	890	6	6
4	4	360	390	0	390	0	390	0	0	390	330	6	6
4	5	890	0	860	0	0	0	800	0	0	800	6	6
4	6	360	330	0	390	0	390	0	0	390	330	6	6
4	7	690	0	860	0	0	0	800	0	0	890	6	6
4	8	890	0	860	0	0	0	800	0	0	890	6	6
5	1	205	205	235	0	0	235	235	205	205	235	9	3
5	2	235	205	205	0	0	235	235	205	235	205	9	3
5	3	0	0	340	310	340	0	0	340	0	310	9	3
5	4	0	0	340	340	340	0	0	340	0	340	9	3
5	5	235	235	235	0	0	205	205	205	235	235	9	3
5	6	0	0	240	340	340	0	0	340	0	340	9	3
5	7	0	0	340	340	340	0	0	240	0	340	9	3
5	8	235	235	235	0	0	235	235	205	235	235	9	3
6	1	265	265	205	235	265	265	0	0	0	265	1	8
6	2	235	235	265	235	265	265	0	0	0	265	1	8
6	3	0	435	105	0	0	0	435	0	435	0	1	8
6	4	0	435	405	0	0	0	435	0	435	0	1	8
6	5	265	265	265	235	265	265	0	0	0	265	1	8
6	6	265	265	265	235	265	265	0	0	0	205	1	8
6	7	0	405	405	0	0	0	435	0	405	0	1	8
6	8	0	265	405	0	0	0	405	0	435	0	1	8
7	1	220	320	0	0	320	0	300	320	300	320	5	5
7	2	525	0	0	525	0	525	0	0	0	325	5	5
7	3	220	320	0	0	320	0	320	320	320	220	5	5
7	4	220	320	0	0	320	0	320	320	320	320	5	5
7	5	525	0	0	525	0	525	0	0	0	505	5	5
7	6	220	220	0	0	320	0	320	320	320	320	5	5
7	7	505	0	0	505	0	525	0	0	0	505	5	5
7	8	525	0	0	525	0	325	0	0	0	505	5	5
8	1	390	330	0	360	360	390	0	0	360	390	4	2
8	2	300	330	0	390	360	390	0	0	390	390	4	2
8	3	0	830	0	0	800	0	0	860	0	0	4	2
8	4	390	330	0	390	360	390	0	0	390	390	4	2
8	5	0	860	0	0	830	0	0	800	0	0	4	2
8	6	390	330	0	390	360	300	0	0	390	390	4	2
8	7	0	800	0	0	830	0	0	860	0	0	4	2
8	8	0	860	0	0	830	0	0	860	0	0	4	2